

PHASE 17 — ECDAT COMMAND CENTER, RISK DASHBOARD, QUANTUM READINESS SCORE & EXECUTIVE DECISION SYSTEM

Project: Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)

Problem Statement: SIH26164

Organization: National Technical Research Organisation (NTRO)

Theme: Blockchain & Cybersecurity

Phase: 17 of 22

Status: Enterprise UX, Risk Visualization, Decision Intelligence & Operational Command Center

1. Purpose of This Phase

At this point, ECDAT has accumulated an enormous amount of intelligence:



But an enterprise security platform cannot simply dump all of this information onto the user.

The user needs to answer:

What is happening?

How dangerous is it?

What should I fix first?

Who owns it?

What happens if I don't fix it?

How much progress have we made?

This phase designs the **ECDAT Command Center**.

2. The Dashboard Is Not Just a UI

A common mistake would be to think:

Dashboard = Charts

ECDAT should instead treat the dashboard as:

A decision-making interface.

The dashboard should convert:

Millions of technical observations

into:

A small number of actionable decisions.

3. The ECDAT Command Center

The main screen should answer four questions immediately:

ECDAT COMMAND CENTER

1. WHAT DO WE HAVE?

Cryptographic Inventory

2. HOW EXPOSED ARE WE?

Quantum Risk

	3. WHAT SHOULD WE DO?	
	Migration Priorities	
	4. HOW FAR HAVE WE COME?	
	Quantum Readiness	

4. User Roles

ECDAT should not show the same dashboard to everyone.

At minimum:

Executive
Security Analyst
Security Architect
Developer
Infrastructure Engineer
Compliance Officer
Administrator

5. Executive View

Executives don't need:

AST nodes
API calls
Source lines

They need:

Risk
Exposure
Business Impact
Migration Progress
Investment Requirements

6. Executive Dashboard

Possible top-level metrics:

- Quantum Readiness Score
- Critical Assets
- High-Risk Assets
- Quantum-Vulnerable Assets
- Migration Progress
- Critical Migration Blockers
- Coverage

7. Executive Example

QUANTUM READINESS

63 / 100

MODERATE

Critical quantum-vulnerable assets: 342

Migration completed: 28%

Critical blockers: 17

Enterprise scan coverage: 91%

8. Security Analyst View

Security analysts need deeper information:

- Asset
- Algorithm
- Evidence
- Risk
- Dependencies
- Certificates
- Libraries
- Configuration
- AI Analysis

9. Developer View

Developers should see:

- Your Applications
- Your Findings
- Affected Files
- Affected Dependencies
- Recommended Changes
- Migration Status

Example:

Payment **API**

3 crypto findings

P1:
ECDSA P-256

P2:
SHA-1 certificate chain

P3:
Legacy **TLS** configuration

10. Infrastructure View

Infrastructure engineers need:

- Hosts
- Containers
- Kubernetes
- Certificates
- TLS**
- HSM**
- KMS**
- Cloud

11. Compliance View

Compliance users need:

- Policy Violations
- Unsupported Algorithms
- Certificate Issues
- Coverage
- Audit Evidence
- Migration Status

12. Main Navigation

A strong ECDAT UI could have:

- Dashboard
- Inventory
- Applications
- Algorithms
- Certificates
- Libraries
- Containers
- Infrastructure
- Risk
- Knowledge Graph
- Migration
- Recommendations
- Policies
- Reports
- Scans
- AI Assistant
- Settings

13. Dashboard Information Hierarchy

Information should follow:

- Enterprise
 - ↓
 - Business Unit
 - ↓
 - Application
 - ↓
 - Cryptographic Asset

↓ Evidence

Users can progressively drill down.

14. Enterprise Overview

First level:

Overall Posture
Risk
Coverage
Migration

15. Business Unit Overview

Example:

Payments
Identity
Infrastructure
HR
Operations

Each receives:

Risk
Assets
Readiness
Migration Progress

16. Application Overview

Example:

Payment API

shows:

Criticality: Critical
Crypto Assets: 23

Quantum Vulnerable: 7

PQC Ready: 4

Migration Blockers: 2

17. Asset Overview

Example:

ECDSA P-256

shows:

Used by:

124 applications

Certificates:

87

Containers:

210

Risk:

High

Quantum Status:

Vulnerable

18. Risk Dashboard

The risk dashboard is one of the most important areas.

It should show:

Risk Distribution

Risk by Business Unit

Risk by Algorithm

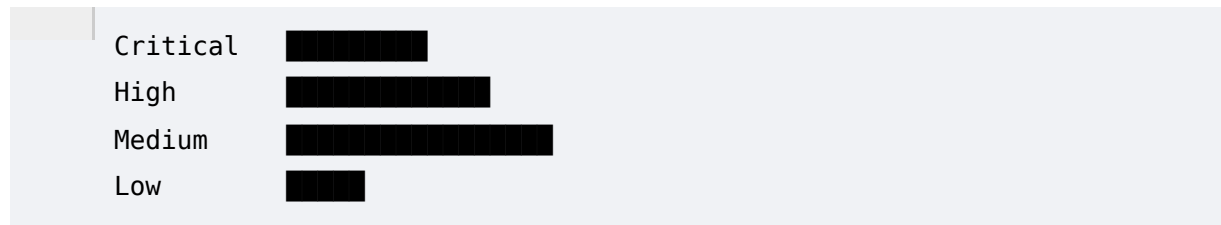
Risk by Application

Risk by Environment

Risk by Asset Type

19. Risk Distribution

Example:



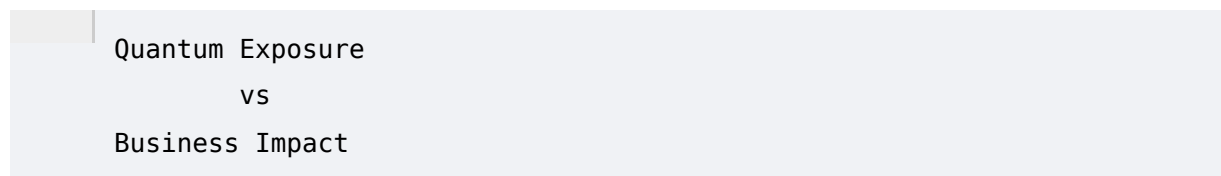
But the actual UI should use a consistent visual language rather than decorative charts.

20. Risk Heatmap

A useful heatmap:



But ECDAT can make the axes more specific:



21. Quantum Risk Matrix

Example:

	Low Business Impact	Medium	High	Critical

Low Quantum Exposure	Low	Low	Medium	Medium
Medium	Low	Medium	High	High
High	Medium	High	Critical	Critical
Extreme	High	Critical	Critical	Critical

This creates intuitive prioritization.

22. Risk Drill-Down

Click:

Critical

and show:

342 assets

Click:

342

and show:

Applications
Algorithms
Certificates
Libraries

Click an application:

Payment API

Then:

Evidence

23. Evidence Panel

The evidence panel should show:

```
Finding
Source
Scanner
Timestamp
File
Line
Configuration
Certificate
Runtime Evidence
Confidence
```

This is essential for trust.

24. "Why Is This Risky?"

Every risk finding should have an explanation.

Example:

```
Why is ECDSA P-256 high risk?
```

ECDAT:

```
Algorithm:
ECDSA

Curve:
P-256

Quantum status:
Vulnerable

Application:
Payment API

Business criticality:
Critical

Data lifetime:
10 years
```

Then:

The system should be prioritized because its data lifetime and estimated migration period create a significant exposure window.

25. Quantum Readiness Score

ECDAT should introduce a high-level metric:

Quantum Readiness Score — QRS.

Scale:

0 → Completely unprepared

100 → Highly prepared

26. Important Warning

The score must **not** be arbitrary.

It should be derived from measurable factors.

Possible dimensions:

Discovery Coverage
Crypto Inventory Quality
Quantum Risk
PQC Adoption
Crypto Agility
Migration Progress
Policy Compliance
Dependency Readiness

27. Example Formula

A conceptual model:

$$QRS = w_cC + w_rR + w_pP + w_aA + w_mM + w_gG$$

where:

- C = coverage
- R = risk posture

- $\backslash(P\backslash)$ = PQC adoption
- $\backslash(A\backslash)$ = crypto agility
- $\backslash(M\backslash)$ = migration progress
- $\backslash(G\backslash)$ = governance

The weights must be configurable.

28. Score Normalization

Each component can be normalized:

0–100

Then:

QRS = weighted combination

This makes the metric explainable.

29. Example

Suppose:

Coverage = 90
 Risk posture = 60
 PQC adoption = 30
 Agility = 70
 Migration = 25
 Governance = 80

The final score might be:

63

The exact number depends on the organization's policy weights.

30. Score Breakdown

Never show only:

63

Show:

Quantum Readiness	
63 / 100	
Coverage	90
Risk Posture	60
PQC Adoption	30
Crypto Agility	70
Migration	25
Governance	80

Now the organization knows what to improve.

31. Readiness Levels

Possible categories:

0–20:
Critical
21–40:
Low
41–60:
Developing
61–80:
Prepared
81–100:
Highly Prepared

These ranges should be configurable.

32. Application Quantum Readiness

The score should not only exist globally.

Each application can have:

Application	QRS
-------------	-----

Example:

Payment API

QRS = 41

Identity Service

QRS = 68

Internal HR

QRS = 82

33. Business Unit Readiness

Example:

Payments:

47

Identity:

38

Infrastructure:

71

HR:

84

This immediately highlights where management should focus.

34. Risk Trend

ECDAT should show risk over time.

Example:

Quantum Risk

Aug 2026: 8,421

Sep 2026: 8,010

Oct 2026: 7,220

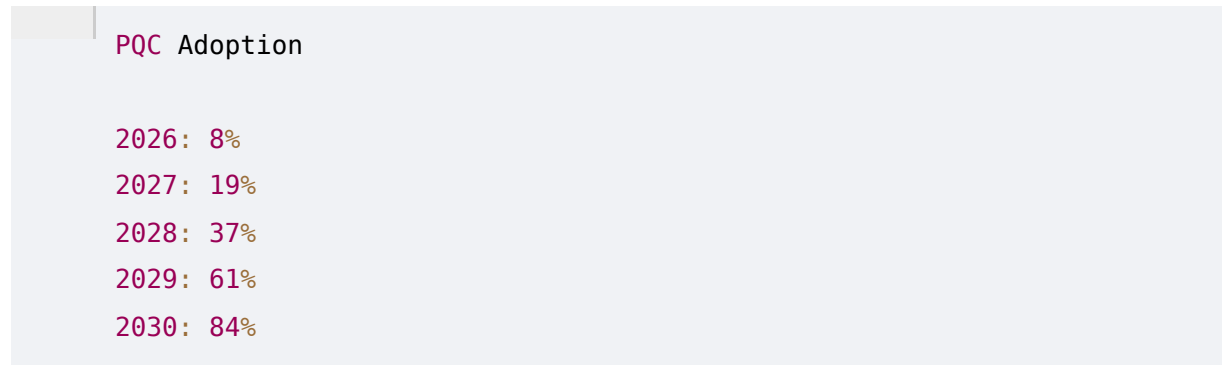
Nov 2026: 6,840

The goal is to show:

Risk declining over time.

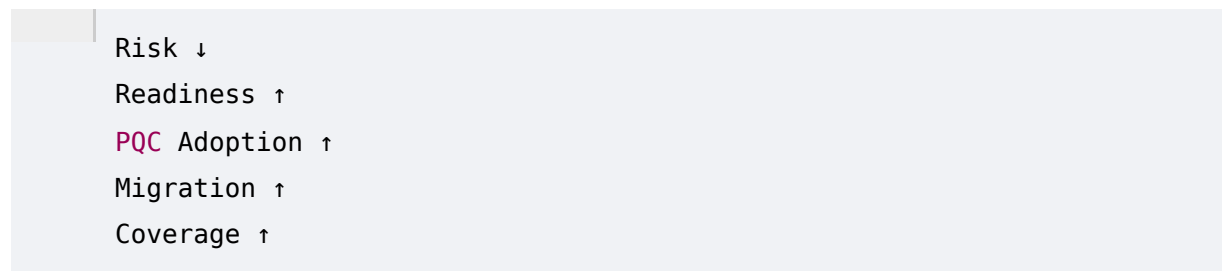
35. Migration Trend

Likewise:



36. Executive Trend View

Executives should see:



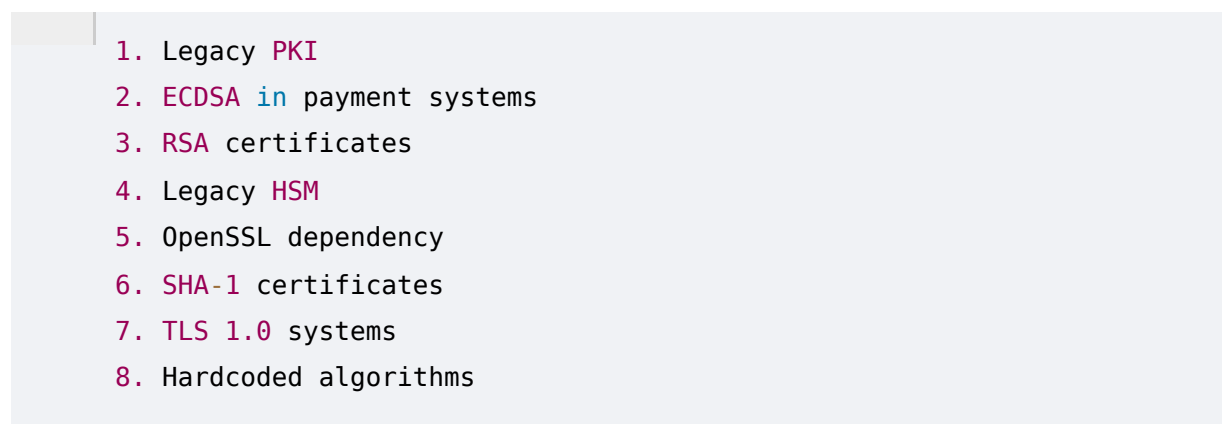
This becomes a transformation dashboard.

37. Top Risks

A dashboard section:

Top 10 Cryptographic Risks

Example:



- 9. Custom cryptography
- 10. Unsupported crypto library

38. Top Migration Blockers

Separate from risk.

Example:

- 1. HSM compatibility
- 2. Legacy client support
- 3. Certificate infrastructure
- 4. Vendor dependency
- 5. Embedded firmware

This distinction is important.

A high-risk asset isn't necessarily the hardest asset to migrate.

39. Risk vs Difficulty Matrix

ECDAT should visualize:

MIGRATION DIFFICULTY			
	Low	Med	High
High Risk	Fix	Plan	Start
Med Risk	Fix	Plan	Plan
Low Risk	Batch	Backlog	Backlog

40. Risk Reduction Opportunities

ECDAT should identify:

Which actions reduce the most risk?

Example:

Upgrade shared crypto provider

could reduce risk across:

2,300 applications

This should rank highly.

41. Migration Opportunity Score

Conceptually:

$$O = \frac{\text{Expected Risk Reduction} \times \text{Blast Radius}}{\text{Estimated Effort}}$$

Again, this is a prioritization metric, not a cryptographic fact.

42. Quick Wins

ECDAT should identify:

High Risk
+
Low Effort

These are:

Quick Wins.

Example:

Replace SHA-1 in internal certificates

43. Strategic Initiatives

Also identify:

High Risk
+
High Effort

These become:

Strategic Migration Programs.

Example:

44. Migration Roadmap

ECDAT should support a roadmap:

2026

Discovery + Inventory

2027

Crypto-agility foundations

2028

Hybrid deployment

2029

PQC migration

2030

Legacy retirement

The dates should be configurable.

45. Application Migration Status

Each application should have:

NOT STARTED

ASSESSED

PLANNED

IN DEVELOPMENT

TESTING

PILOT

MIGRATED

VERIFIED

RETIRED

46. Migration Board

A Kanban-like view:

Assessed	Planned	In Progress	Verified
App A	App D	App G	App J
App B	App E	App H	App K
App C	App F	App I	App L

47. Migration Program View

A migration initiative can contain:

- Goal
- Owner
- Applications
- Dependencies
- Milestones
- Risk
- Status

48. Migration Milestones

Example:

- ☒ Inventory
- ☒ Risk assessment
- ☒ Architecture
- ☐ **PQC** pilot
- ☐ Compatibility testing
- ☐ Production rollout
- ☐ Legacy retirement

49. Verification

Migration is not complete simply because:

Developer says:
"We migrated."

ECDAT should rescan.

```
Migration
↓
Rescan
↓
Compare Graph
↓
Verify Old Crypto Removed
↓
Verify New Crypto Present
```

50. Migration Verification Dashboard

Example:

```
Migration:
Payment API

Expected:
ECDSA → Hybrid

Observed:
ECDSA + ML-DSA

Status:
VERIFIED
```

51. Policy Dashboard

ECDAT should maintain cryptographic policies.

Examples:

```
RSA < 2048 → Forbidden
SHA-1 → Forbidden
TLS 1.0 → Forbidden
Custom crypto → Review
Approved PQC → Allowed
```

52. Policy Violation Dashboard

Show:

```
Critical violations
High violations
Medium violations
Resolved
Unresolved
```

53. Policy-as-Code

Policies should be machine-readable.

Conceptually:

```
algorithm: SHA-1
status: forbidden
scope: production
```

This enables automated assessment.

54. Policy Enforcement

Policies can operate at:

```
Discovery
CI/CD
Runtime
Migration
Reporting
```

55. CI/CD Gate

Developer pushes:

```
New RSA-1024 usage
```

ECDAT:

```
Policy violation
↓
Build blocked
```

depending on organization policy.

56. Alerting

ECDAT should support alerts for:

```
New high-risk asset
New forbidden algorithm
Certificate expiration
Crypto drift
New vulnerable library
Migration regression
Coverage degradation
```

57. Alert Severity

```
INFO
WARNING
HIGH
CRITICAL
```

58. Alert Example

```
CRITICAL

New quantum-vulnerable cryptographic asset
detected in production.

Application:
Payment API

Algorithm:
RSA-1024

Owner:
Payments Team
```

59. Notification Channels

Potential integrations:

```
Email
Slack
Microsoft Teams
SIEM
Webhook
Ticketing System
```

Actual integrations depend on deployment.

60. Ticket Creation

ECDAT can generate remediation tickets.

Example:

```
Title:
Migrate ECDSA P-256 in Payment API

Priority:
P1

Owner:
Payments Team

Evidence:
...

Recommendation:
...
```

61. SIEM Integration

ECDAT should emit structured events:

```
{
  "event": "CRYPTO_RISK_DETECTED",
  "severity": "high",
  "application": "payment-api",
  "algorithm": "ECDSA"
}
```

This allows SOC integration.

62. Report Generation

ECDAT should produce several report types.

Executive

1–5 pages

Security

Detailed risk

Technical

Evidence
Files
Dependencies

Compliance

Policies
Violations
Audit evidence

Migration

Roadmap
Progress
Blockers

63. Executive Report

Possible sections:

Executive Summary
Quantum Readiness
Top Risks
Business Impact
Migration Progress
Top Blockers
Recommended Actions

64. Technical Report

Include:

- Asset
- Algorithm
- Parameters
- Library
- Version
- Evidence
- Risk
- Recommendation
- Dependencies

65. Report Traceability

Every major statement should be traceable.

Example:

"342 critical assets"

should link to:

- Query
- Timestamp
- Dataset Version

66. Auditability

ECDAT should log:

- Who
- Did what
- When
- Against which asset
- Using which evidence
- Result

67. Search

The dashboard should include global search.

User types:

	RSA
--	-----

Results:

	Algorithms
	Applications
	Certificates
	Libraries
	Findings
	Reports

68. Natural-Language Search

Powered by Phase 14:

"Show all critical applications using quantum-vulnerable signatures."

ECDAT generates a structured query.

69. Saved Queries

Users can save:

	Critical PQC blockers
	Quantum-vulnerable production
	Expiring certificates
	Policy violations

70. Custom Dashboards

Organizations may want custom views.

Example:

	National Infrastructure Dashboard
--	-----------------------------------

with:

	Critical Systems
	PQC Readiness
	Certificate Health

71. Dashboard Widgets

Potential widgets:

Readiness Score
Risk Score
Asset Count
Coverage
Top Algorithms
Top Libraries
Top Risks
Migration Progress
Blast Radius
Policy Violations

72. Drill-Down Principle

Every number should be clickable.

Example:

342 Critical Assets

click →

List of 342

click asset →

Asset Detail

click evidence →

Source

This is:

Progressive Disclosure.

73. Asset Detail Page

Example:

ECDSA P-256
Status:
Quantum Vulnerable
Risk:
High
Applications:
124
Certificates:
87
Libraries:
6
Business Processes:

74. Evidence Timeline

Show:

Aug 01:
Detected
Aug 04:
Confirmed
Aug 08:
Risk assessed
Aug 15:
Migration planned
Aug 20:
Migration started

75. Asset History

This allows analysts to understand:

What changed?
When?
Who changed it?
Did risk increase?
Did risk decrease?

76. Risk Explainability

Every risk score should have:

Risk Score
+
Contributing Factors
+
Evidence
+
Business Context

Example:

Risk: 87

Factors:

Quantum vulnerability	+30
Critical application	+25
Long-lived data	+20
Migration complexity	+12
Existing mitigation	-5

77. Risk Factor Transparency

Do not show:

Risk = 87

without explanation.

Show:

Why 87?

78. Executive "Why?"

Executives should be able to ask:

Why is our readiness score only 63?

ECDAT:

Primary reasons:

1. High quantum exposure
2. Low PQC adoption
3. Legacy PKI
4. Crypto-agility gaps
5. Migration blockers

79. Executive "What Next?"

ECDAT can answer:

What should we do next?

Example:

1. Address legacy PKI
2. Upgrade shared crypto providers
3. Pilot hybrid signatures
4. Migrate critical applications
5. Verify migration through rescanning

80. Decision Intelligence

This is the final layer.

ECDAT should move from:

Monitoring

to:

Decision Support

81. Recommended Action Cards

Example:

PRIORITY ACTION
Upgrade legacy PKI
Risk reduction: HIGH
Blast radius: CRITICAL
Effort: HIGH
120 applications affected
[VIEW IMPACT] [CREATE PLAN]

82. Top Strategic Recommendations

ECDAT should generate a ranked list:

1. Migrate legacy **PKI**
2. Upgrade crypto provider
3. Introduce crypto abstraction
4. Pilot hybrid **PQC**
5. Retire weak certificates

83. Risk Acceptance

Not every risk can immediately be fixed.

ECDAT should support:

Accept
Mitigate
Transfer
Avoid
Defer

84. Risk Acceptance Workflow

Example:

Finding

↓

Review

↓

Business justification

↓

Approver

↓

Expiration date

↓

Accepted temporarily

Risk acceptance should never mean:

Delete finding

85. Temporary Exceptions

Example:

RSA usage

Exception:

Legacy vendor compatibility

Approved by:

Security Architecture

Expires:

2027-03-01

ECDAT alerts before expiration.

86. Exception Dashboard

Show:

Active Exceptions

Expiring Soon

Expired

High-Risk Exceptions

87. Multi-Tenant Architecture

If ECDAT becomes an enterprise SaaS/platform:

```
Organization
├── Business Unit
│   ├── Applications
│   └── Assets
```

Tenant isolation becomes critical.

88. Organization Isolation

No tenant should be able to query another tenant's:

```
Assets
Applications
Graph
Reports
Evidence
```

89. Dashboard Performance

Large organizations may have:

```
Millions of assets
```

Therefore dashboards should use:

```
Aggregations
Caching
Precomputed metrics
Pagination
Async queries
```

90. Real-Time vs Batch

Not everything needs real-time updates.

Real-time

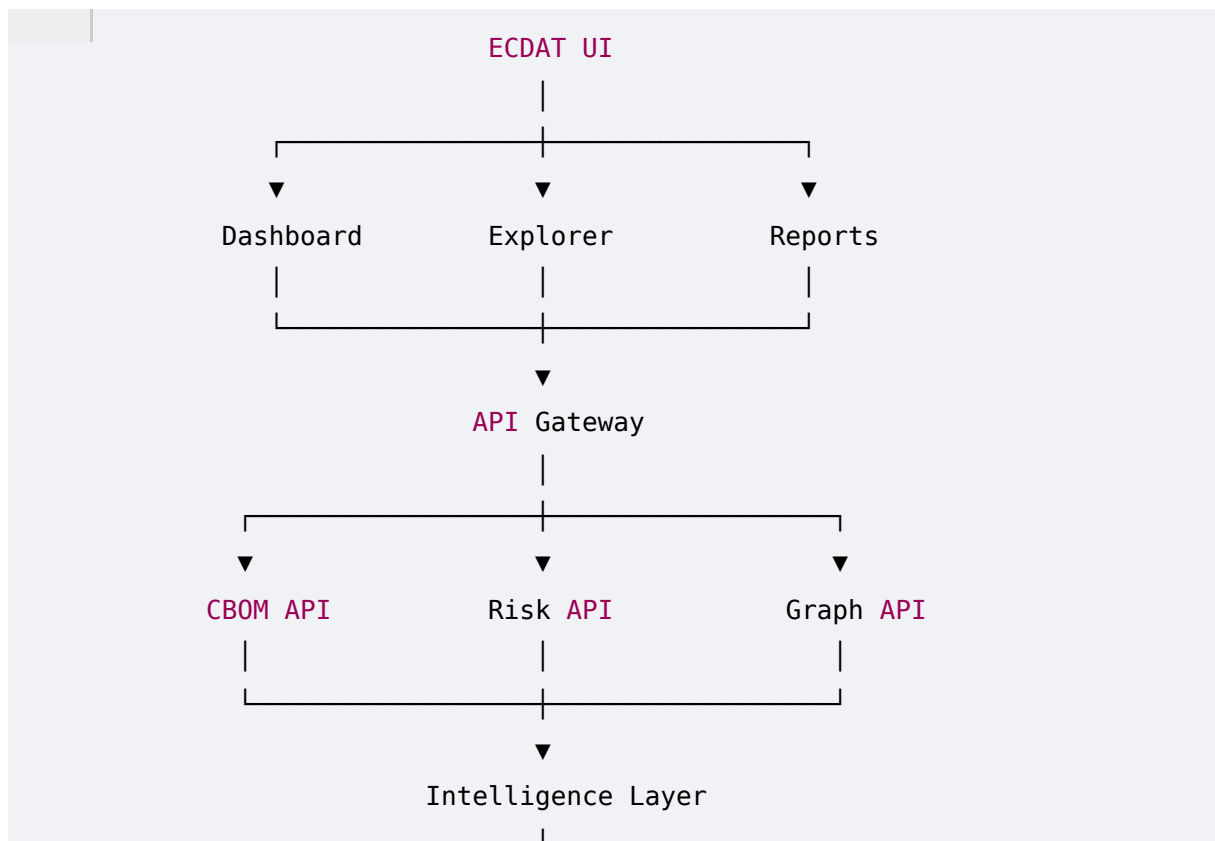
```
Critical alerts
New findings
```

Scan status

Batch

Readiness score
Centrality
Portfolio analytics
Historical reports

91. Command Center Architecture



92. Frontend Architecture

Potential implementation:

React
+
TypeScript

with:

Component Library
Charts

Graph Visualization
Tables
Filters
Search

93. Backend Architecture

Potential:

```
graph TD; A[API Gateway] --> B[Services]; B --> C[CBOM]; C --> D[Risk]; D --> E[Graph]; E --> F[Scan]; F --> G[Recommendation]; G --> H[AI];
```

API Gateway
↓
Services
↓
CBOM
Risk
Graph
Scan
Recommendation
AI

A modular monolith can be preferable for the SIH prototype.

94. SIH Prototype Recommendation

Do **not** build 20 microservices for the hackathon.

Build:

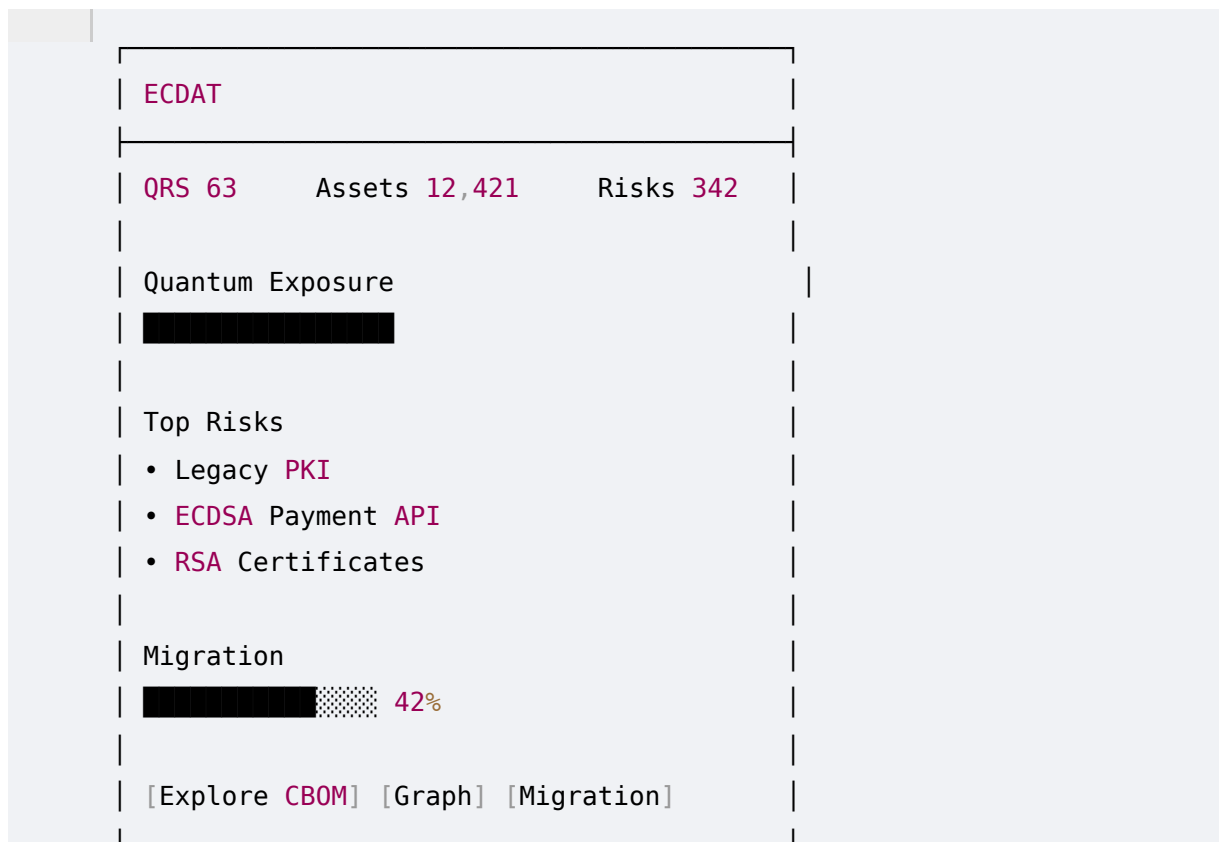
```
graph TD; A[Frontend] --> B[Backend]; B --> C[PostgreSQL]; C --> D[Graph Layer]; D --> E[Scanner Workers];
```

Frontend
↓
Backend
↓
PostgreSQL
↓
Graph Layer
↓
Scanner Workers

Then demonstrate extensibility.

95. Prototype Dashboard

A strong SIH dashboard could contain:

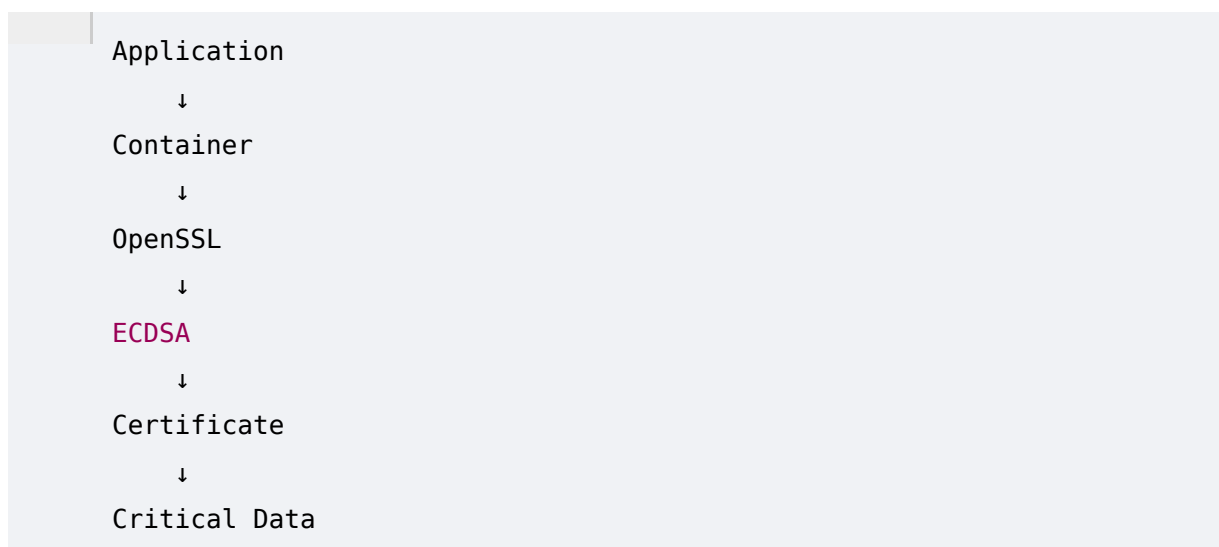


96. Killer UI Interaction

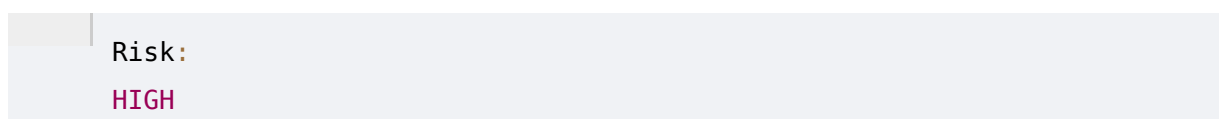
Click:



Then animate:



Then:



Blast Radius:

MEDIUM

Recommendation:

Hybrid migration

This ties together almost every previous phase.

97. Command Center's Main Principle

The UI should answer:

WHAT?

WHY?

SO WHAT?

NOW WHAT?

WHAT?

What cryptography exists?

WHY?

Why is it risky?

SO WHAT?

What business systems are affected?

NOW WHAT?

What should we do?

98. The Complete Decision Loop

DISCOVERY



CBOM



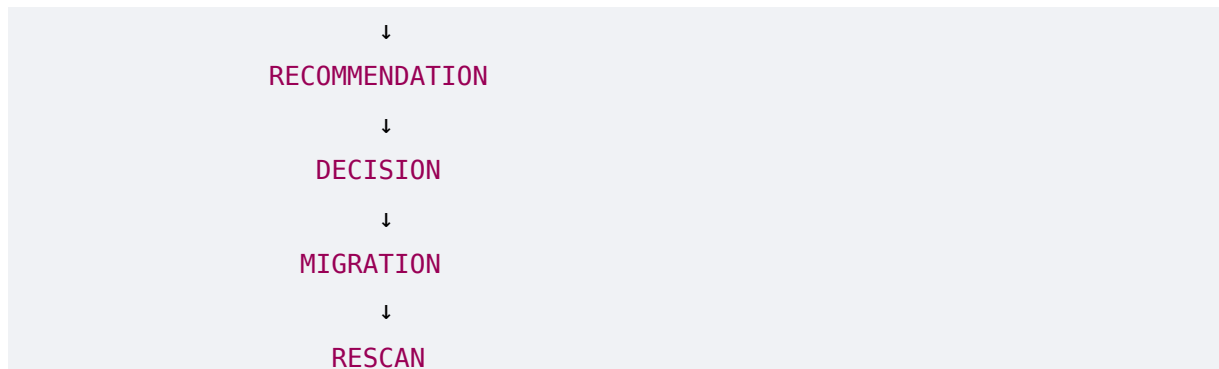
GRAPH



RISK



BUSINESS IMPACT



This loop should be visible throughout the product.

99. Phase 17 Requirements

ECDAT Command Center should provide:

Executive

- Quantum Readiness Score
- Enterprise Risk
- Migration Progress
- Top Risks
- Top Blockers
- Business Impact

Security

- Asset Explorer
- Evidence
- Risk
- Graph
- Alerts
- Policies

Developers

- Findings
- Source Locations
- Dependencies
- Recommendations

Infrastructure

- Containers

- Certificates
- TLS
- HSM
- KMS
- Cloud

Management

- Migration roadmap
- Investment priorities
- Risk reduction
- Exceptions
- KPIs

100. Core KPIs

ECDAT should track:



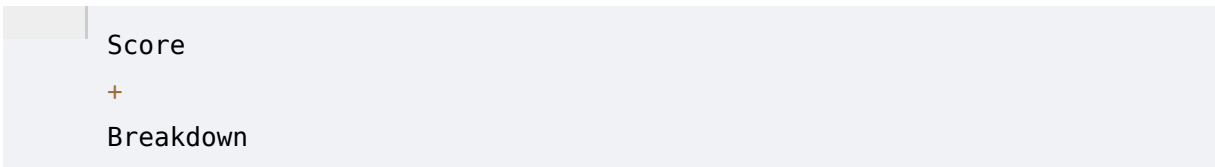
- Total Crypto Assets
- Quantum Vulnerable Assets
- Critical Quantum Assets
- PQC Adoption
- Hybrid Adoption
- Crypto Agility
- Migration Progress
- Coverage
- Policy Violations
- Migration Blockers
- Blast Radius
- Risk Reduction

101. Most Important KPI

The strongest single executive KPI is:

Quantum Readiness Score.

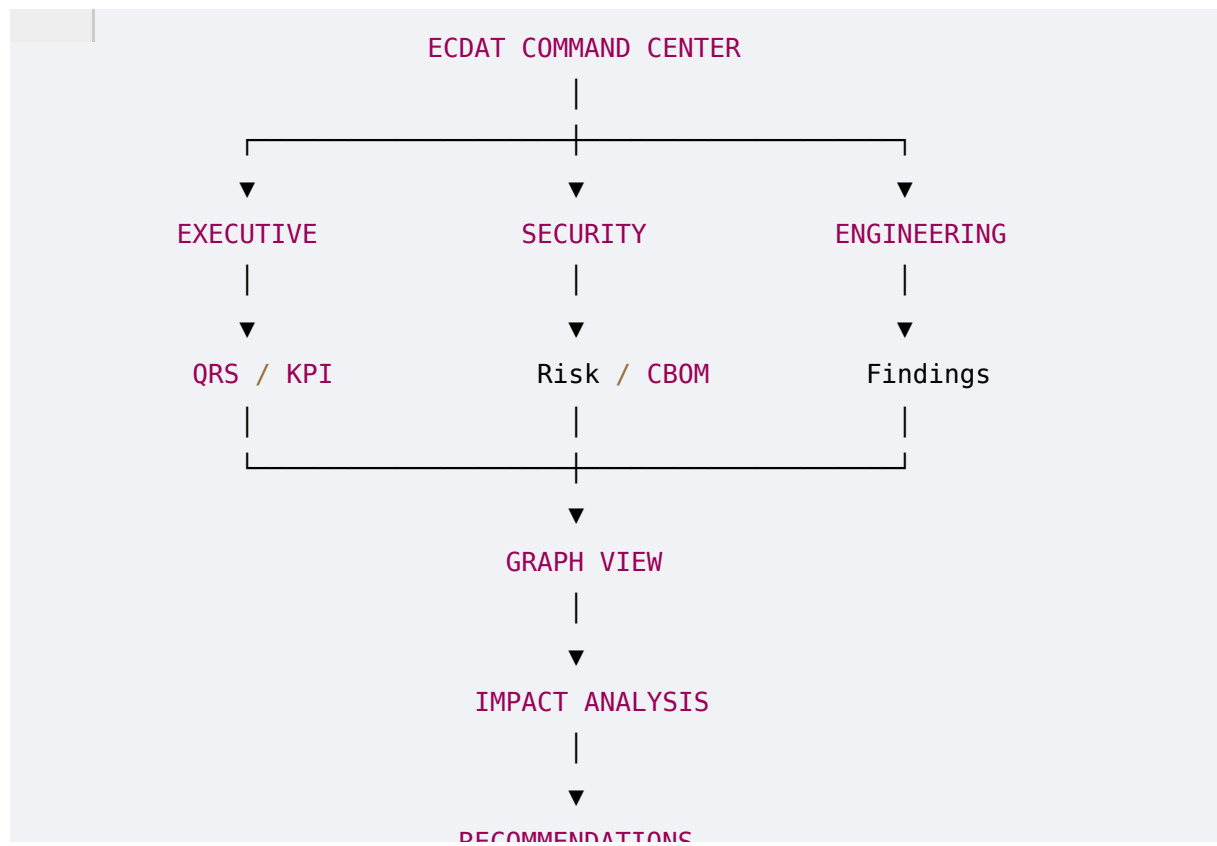
But the system must always provide:



- Score
- +
- Breakdown

- + Evidence
- + Trend
- + Recommended Actions

102. Phase 17 Final Architecture



103. Phase 17 Final Takeaway

The purpose of ECDAT's dashboard is **not** to make the product look sophisticated.

The dashboard exists to convert:

Technical complexity

into:

Actionable decisions.

The complete experience should therefore be:

I discovered a cryptographic asset.



I know exactly where it is.



I know what it protects.



I know how risky it is.



I know what depends on it.



I know what happens if I change it.



I know what I should migrate to.



I know who owns the migration.



I can track the migration.

That is the difference between a **dashboard** and an **enterprise cryptographic command center**.

PHASE 18 PREVIEW — CRYPTO-AGILITY, PQC/HYBRID MIGRATION PLANNER & AUTOMATED TRANSITION ENGINE

The next phase moves from:

"We know what is wrong."

to:

"Let's actually plan how to fix it."

Phase 18 will design the complete migration engine:

Current Crypto

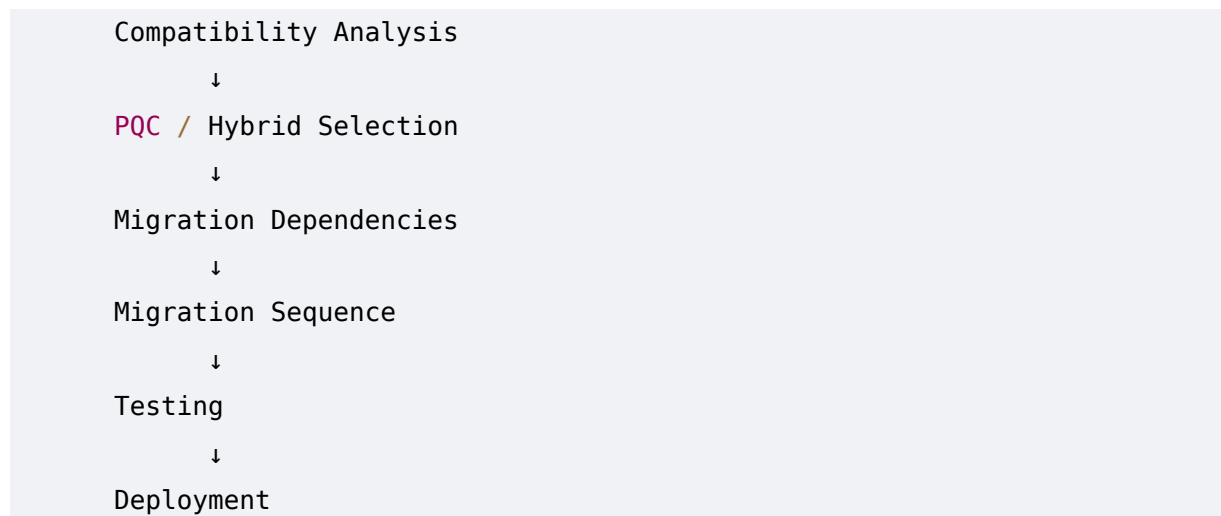


Risk



Target State



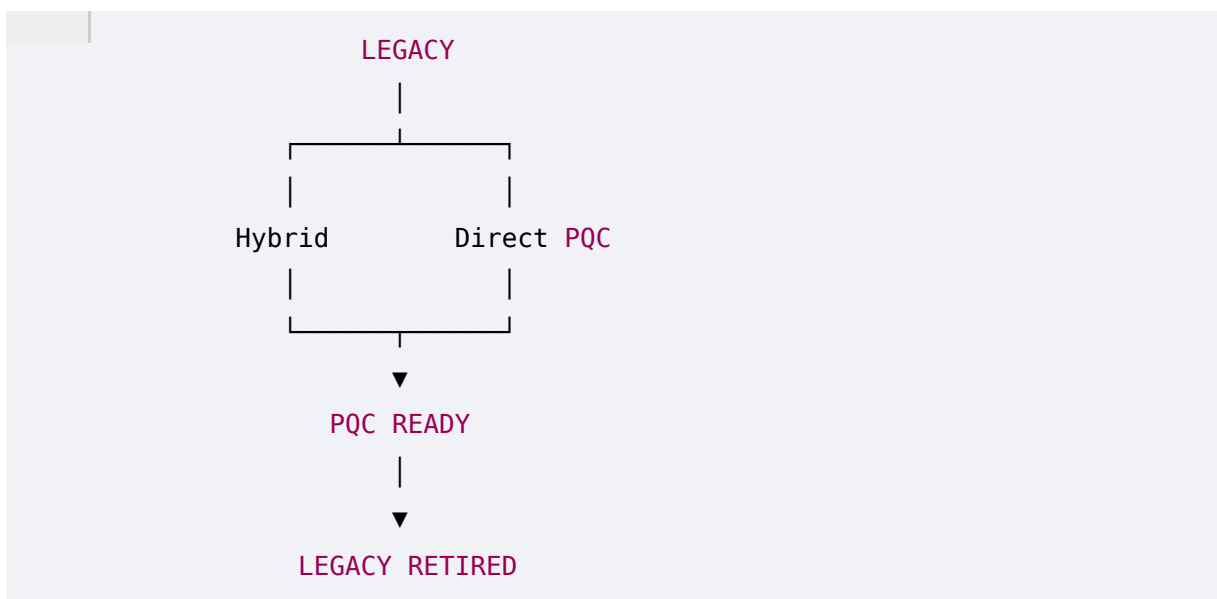


It will cover:

- Crypto-agility architecture
- Algorithm abstraction
- Provider abstraction
- Configuration-driven crypto
- PQC integration
- Hybrid cryptography
- Algorithm negotiation
- Protocol compatibility
- Certificate migration
- PKI migration
- Key migration
- HSM/KMS migration
- Application migration
- Library migration
- Container migration
- API compatibility
- Backward compatibility
- Legacy clients
- Rollback
- Canary deployments
- Dual-sign / dual-encrypt strategies
- Hybrid certificates where applicable
- Migration waves

- Dependency ordering
- Migration critical paths
- Automated migration planning
- Migration effort estimation
- Risk reduction estimation
- Testing strategy
- Performance benchmarking
- Regression detection
- Migration readiness gates
- Post-migration verification
- Crypto-agility scoring
- Long-term algorithm replacement strategy

The central concept will be:



The goal is not simply to recommend **a new algorithm**.

It is to design a **safe, measurable, dependency-aware transition from today's cryptographic infrastructure to a quantum-resistant future**.

Phase 17 complete.